Question 1

Question 1

1. Weekday $W=0$; $T=0, Y=1 : 0.052$; $T=1, Y=1 : 0.158$

   Weekend $W=1$: $T=0, Y=1 : 0.036$; $T=1, Y=1 : 0.144$

   $P(Y=1) = 0.052 + 0.158 + 0.036 + 0.144 = 0.390$

   $P(Y=1) = 0.39$

2. $P(Y=1 \mid T=1) = \dfrac{P(Y=1 \text{ and } T=1)}{P(T=1)}$

   $P(Y=1, T=1) = 0.158 + 0.144 = 0.302$

   $P(T=1) = 0.074 + 0.158 + 0.030 + 0.144 = 0.406$

   $P(Y=1 \mid T=1) = \dfrac{0.302}{0.406} = 0.744$

3. $P(T=1 \mid Y=1) = \dfrac{P(T=1 \text{ and } Y=1)}{P(Y=1)}$

   $P(T=1, Y=1) = 0.144 + 0.158 = 0.302$

   $P(Y=1) = 0.390$ (1.1)

   $P(T=1 \mid Y=1) = \dfrac{0.302}{0.390} = 0.774$

4. They are different conditional probabilities.

   2 asks: when the system forecasts high demand, how often is it correct?

   3 asks: when high demand really occurs, how often does the system detect it?

   different denominators, so they are not generally equal.

5. $P(\text{system correct} \mid W=0) = \dfrac{P(\text{system correct and } W=0)}{P(W=0)}$

   2 system correct:

   ① $W=0, T=0, Y=0 : 0.416$

   ② $W=0, T=1, Y=1 : 0.158$

   $P(\text{system correct and } W=0) = 0.416 + 0.158 = 0.574$

   $P(W=0) = 0.416 + 0.052 + 0.074 + 0.158 = 0.700$

   $P(\text{system correct} \mid W=0) = \dfrac{0.574}{0.700} = 0.820$

   $P(\text{system correct} \mid W=0) = 0.82$

6. $P(\text{system correct} \mid W=1) = \dfrac{P(\text{system correct and } W=1)}{P(W=1)}$

   ① $W=1, T=0, Y=0 : 0.090$

   ② $W=1, T=1, Y=1 : 0.144$

   $P(\text{system correct and } W=1) = 0.090 + 0.144 = 0.234$

   $P(W=1) = 0.090 + 0.036 + 0.030 + 0.144 = 0.300$

   $P(\text{system correct} \mid W=1) = \dfrac{0.234}{0.300} = 0.780.$

   $P(\text{system correct} \mid W=1) = 0.78.$

Question 2

Question 2

1. $\hat{\tau}_w = w\hat{\tau}_1 + (1-w)\hat{\tau}_2$

$\hat{\tau}_1 = \tau + e_1 \quad \hat{\tau}_2 = \tau + e_2$

so $\hat{\tau}_w = w(\tau+e_1) + (1-w)(\tau+e_2)$

$= w\tau + we_1 + (1-w)\tau + (1-w)e_2$

$= \tau + we_1 + (1-w)e_2$

$\hat{\tau}_w - \tau = we_1 + (1-w)e_2$

$E[\hat{\tau}_w - \tau] = E[we_1 + (1-w)e_2]$

$= wE[e_1] + (1-w)E[e_2]$

because $E[e_1] = E[e_2] = 0$

so $E[\hat{\tau}_w - \tau] = 0$

~~therefore $\hat{\tau}_w$~~

2. we know $\hat{\tau}_w - \tau = we_1 + (1-w)e_2$ in 2.1

so $E[(\hat{\tau}_w - \tau)^2] = E[(we_1 + (1-w)e_2)^2]$

$(we_1 + (1-w)e_2)^2 = w^2e_1^2 + (1-w)^2e_2^2 + 2w(1-w)e_1e_2$

so $E[(\hat{\tau}_w - \tau)^2] = w^2E[e_1^2] + (1-w)^2E[e_2^2] + 2w(1-w)E[e_1e_2]$

$E[e_1e_2] = E[e_1]E[e_2] = 0$ so $2w(1-w)E[e_1e_2] = 0$

because $V[e_1] = 16 \quad V[e_2] = 9$

so $E[e_1^2] = 16 \quad E[e_2^2] = 9$

$E[(\hat{\tau}_w - \tau)^2] = 16w^2 + 9(1-w)^2$

$= 16w^2 + 9(1 - 2w + w^2)$

$= 16w^2 + 9 - 18w + 9w^2$

$= 25w^2 - 18w + 9$

$E[(\hat{\tau}_w - \tau)^2] = 25w^2 - 18w + 9$

3. $\frac{d}{dw}(25w^2 - 18w + 9) = 50w - 18$

let $50w - 18 = 0$

$50w = 18$

$w = \frac{18}{50}$

$w = 0.36$

4. $E[(Fw-D)^2] = 16w^2 + 9(1-w)^2$

$w = 0.36$

$MSFE = 16w^2 + 9(1-w)^2$

$= 16(0.36)^2 + 9(1-0.36)^2$

$= 16(0.1296) + 9(0.4096)$

$= 2.0736 + 3.6864$

$= 5.76$

use F_1 alone:

$MSFE = 16(1)^2 + 9(1-1)^2$

$= 16 + 0 = 16$

use F_2 alone:

$MSFE = 16(0)^2 + 9(1-0)^2$

$= 0 + 9 = 9$

simple average:

$(F_1 + F_2)/2$ means $w = 0.5$

$MSFE = 16(0.5)^2 + 9(0.5)^2$

$= 16(0.25) + 9(0.25)$

$= 4 + 2.25$

$= 6.25$

F_1 alone: 16

F_2 alone: 9

Simple average: 6.25

Optimal combo: 5.76

$5.76 < 6.25 < 9 < 16$

Therefore, the optimal combined forecast has the smallest MSFE and performs

5. $E[f(x)] \approx f(\mu x) + \frac{\sigma^2 x}{2} f''(\mu x)$

$E[C(D)] \approx C(E[D]) + \frac{1}{2} C''(E[D]) V[D]$

We know $E[D] = 30$ $V[D] = 25$

$E[C(D)] \approx C(30) + \frac{1}{2} C''(30)(25)$

we know $C(D) = \frac{1000}{D+10}$

let $D = 30$

$C(30) = \frac{1000}{30+10} = 25$

$E[C(D)] \approx 25 + \frac{1}{2} C''(30)(25)$

$C(D) = \frac{1000}{D+10}$

$\frac{1}{D+10} = (D+10)^{-1}$

$C(D) = 1000(D+10)^{-1}$

$C'(D) = 1000(-1)(D+10)^{(-1-1)}$

$= -1000(D+10)^{-2}$

$C''(D) = -1000(-2)(D+10)^{(-2-1)}$

$= 2000(D+10)^{-3}$

$= \frac{2000}{(D+10)^3}$

Let $D = 30$

$C''(30) = \frac{2000}{(30+10)^3} = \frac{2000}{40^3} = \frac{2000}{64000} = \frac{1}{32} = 0.03125$

$C''(30) = 0.03125$

$E[C(D)] \approx 25 + \frac{1}{2}(0.03125)(25)$

$\approx 25 + 0.390625$

≈ 25.390625

Hence, $E[C(D)] \approx \$25.39$

6. $V[f(X)] = (f'(\mu_X))^2 \sigma_X^2$

~~$V[C(D)] \approx (C'(E(C)))^2 \sigma_X^2$~~

$V[C(D)] \approx (C'(E(C)))^2 \cdot V[D]$

$V[C(D)] \approx C'(30)^2 (25)$

$C'(D) = -1000(D+10)^{-2}$ (we know from 2.5)

$= -\frac{1000}{(D+10)^2}$

Let $D = 30$

$C'(30) = -\frac{1000}{(30+10)^2} = -\frac{1000}{40^2} = -\frac{1000}{1600} = -\frac{5}{8} = -0.625$

So. $V[C(D)] \approx (-0.625)^2 (25)$

$\approx 0.390625(25)$

≈ 9.765625

Hence, $V[C(D)] \approx$ ~~9.765~~ 9.77

Question 3

Question 3

1. $f(x) = 1 + \theta(2x-1)$

case 1 : $\theta = -1$

$f(x) = 1 + (-1)(2x-1)$

$= 1 - 2x + 1$

$= 2 - 2x$

$0 \le x \le 1$

when $x = 0$ $f(0) = 2$

when $x = 1$ $f(1) = 0$

case 2 : $\theta = 0$

$f(x) = 1 + 0(2x-1)$

$= 1$

case 3 : $\theta = 1$

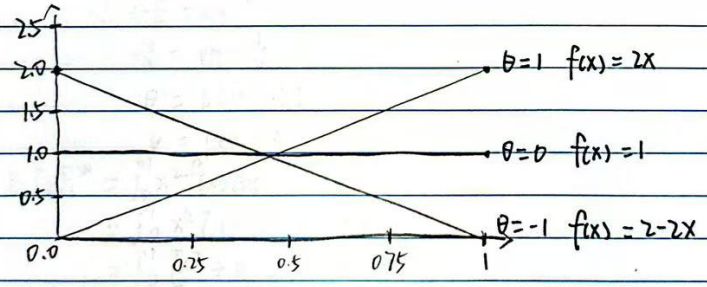
$f(x) = 1 + 1(2x-1)$

$= 1 + 2x - 1$

$= 2x$

when $x = 0$ $f(0) = 0$

when $x = 1$ $f(1) = 2$



2. valid PDF $\begin{cases} f(x) \ge 0 \\ \int_{-\infty}^{\infty} f(x)dx = 1 \end{cases}$

① $f(x) = 1 + \theta(2x-1)$

because $0 \le x \le 1$

so $-1 \le 2x-1 \le 1$

because $-1 \le \theta \le 1$

so $\theta(2x-1) \ge -1$

so $f(x) = 1 + \theta(2x-1) \ge 0$

$f(x) \ge 0$

② because $0 \le x \le 1$

so $\int_0^1 [1+\theta(2x-1)]\,dx$

$= \int_0^1 1\,dx + \theta\int_0^1 (2x-1)\,dx$

$= 1 + \theta[x^2-x]_0^1$

$= 1 + \theta(1^2-1) - (0^2-0)$

$= 1$

therefore $\int_0^1 f(x)\,dx = 1$

Hence, this is a valid PDF.

3. $E[X] = \int_0^1 x f(x)\,dx$

$= \int_0^1 x[1+\theta(2x-1)]\,dx$

$= \int_0^1 [x+\theta(2x^2-x)]\,dx$

$= \int_0^1 x\,dx + \theta\int_0^1 (2x^2-x)\,dx$

$= [\frac{x^2}{2}]_0^1 + \theta[\frac{2x^3}{3} - \frac{x^2}{2}]_0^1$

$= \frac{1}{2} + \theta(\frac{2}{3} - \frac{1}{2})$

$= \frac{1}{2} + \frac{\theta}{6}$

Hence, $E[X] = \frac{1}{2} + \frac{\theta}{6}$

4. Let $E[X] = m$

we know $E[X] = \frac{1}{2} + \frac{\theta}{6}$ at 3.3.

$\frac{1}{2} + \frac{\theta}{6} = m$

$\frac{\theta}{6} = m - \frac{1}{2}$

$\theta = 6(m - \frac{1}{2})$

$\theta = 6m - 3$

5. $E[X^2] = \int_0^1 x^2 f(x)\,dx$

$= \int_0^1 x^2[1+\theta(2x-1)]\,dx$

$= \int_0^1 [x^2 + \theta(2x^3 - x^2)]\,dx$

$= \int_0^1 x^2\,dx + \theta\int_0^1 (2x^3 - x^2)\,dx$

$= [\frac{x^3}{3}]_0^1 + \theta[\frac{x^4}{2} - \frac{x^3}{3}]_0^1$

$= \frac{1}{3} + \theta(\frac{1}{2} - \frac{1}{3})$

$= \frac{1}{3} + \frac{\theta}{6}$

Hence, $E[X^2] = \frac{1}{3} + \frac{\theta}{6}$

6. $V[X] = E[X^2] - (E[X])^2$

$= \frac{1}{3} + \frac{\theta}{6} - (\frac{1}{2} + \frac{\theta}{6})^2$

$= \frac{1}{3} + \frac{\theta}{6} - (\frac{1}{4} + \frac{\theta}{6} + \frac{\theta^2}{36})$

$= \frac{1}{12} - \frac{\theta^2}{36}$ Hence, $V[X] = \frac{1}{12} - \frac{\theta^2}{36}$

7. CDF means $F(x_0) = P(X \le x_0)$

$F(x_0) = \int_{-\infty}^{x_0} f(x)\,dx$

because $0 \le X \le 1$

so if $0 \le x_0 \le 1$

then $F(x_0) = \int_0^{x_0} [1 + \theta(2x-1)]\,dx$

$= \int_0^{x_0} 1\,dx + \int_0^{x_0} \theta(2x-1)\,dx$

$= [x]_0^{x_0} + [\theta(x^2 - x)]_0^{x_0}$

$= x_0 + \theta(x_0^2 - x_0)$

Hence, $F(x_0) = x_0 + \theta(x_0^2 - x_0)$

CDF 3 cases:

① $x_0 < 0$, but $X \ge 0$, so $P(X \le x_0) = 0$

$F(x_0) = 0$

② ~~$x_0 = 0$~~ $0 \le x_0 \le 1$

$F(x_0) = x_0 + \theta(x_0^2 - x_0)$

③ $x_0 > 1$, ~~but~~ because $X \le 1$, so $P(X \le x_0) = 1$

$F(x_0) = 1$

Hence, $F(x_0) \begin{cases} 0, & x_0 < 0 \\ x_0 + \theta(x_0^2 - x_0), & 0 \le x_0 \le 1 \\ 1, & x_0 > 1 \end{cases}$

8. median means $P(X \le m) = 0.5$

$F(m) = 0.5$

turn $\theta = 1$ into $F(x) = x + \theta(x^2 - x)$

$F(x) = x + x^2 - x = x^2$

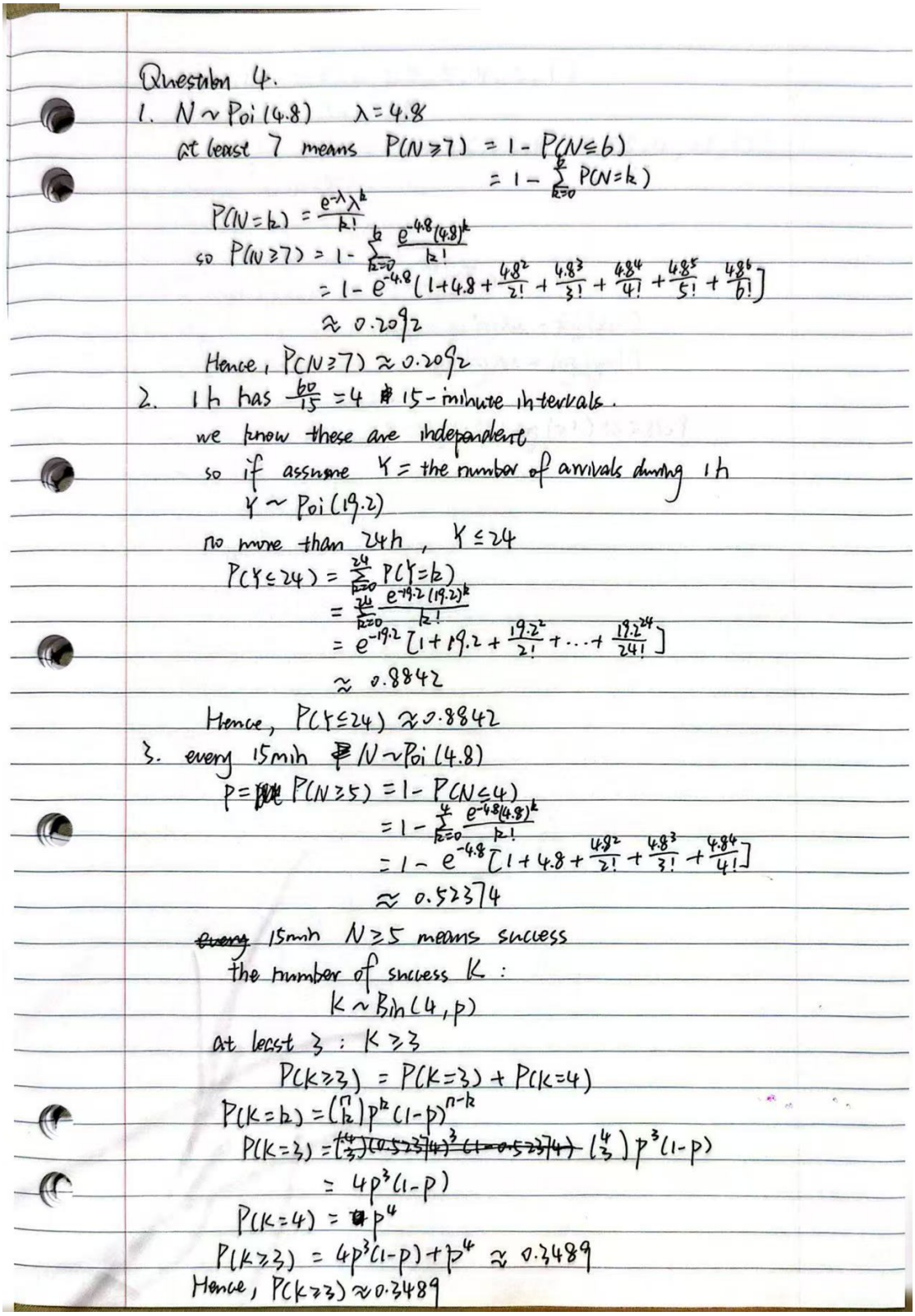
let $x = m$

$F(m) = m^2 = 0.5$

$m = \sqrt{0.5} = \frac{1}{\sqrt{2}} \approx 0.7071$

Hence, median $= \frac{1}{\sqrt{2}} \approx 0.707$

Question 4



Question 4.

1. $N \sim Poi(4.8)$ $\lambda = 4.8$

at least 7 means $P(N \geq 7) = 1 - P(N \leq 6)$

$$= 1 - \sum_{k=0}^{k} P(N=k)$$

$$P(N=k) = \frac{e^{-\lambda}\lambda^k}{k!}$$

so $P(N \geq 7) = 1 - \sum_{k=0}^{6} \frac{e^{-4.8}(4.8)^k}{k!}$

$$= 1 - e^{-4.8}\left(1 + 4.8 + \frac{4.8^2}{2!} + \frac{4.8^3}{3!} + \frac{4.8^4}{4!} + \frac{4.8^5}{5!} + \frac{4.8^6}{6!}\right)$$

$$\approx 0.2092$$

Hence, $P(N \geq 7) \approx 0.2092$

2. 1h has $\frac{60}{15} = 4$ 个 15-minute intervals.

we know these are independent

so if assume Y = the number of arrivals during 1h

$Y \sim Poi(19.2)$

no more than 24h, $Y \leq 24$

$$P(Y \leq 24) = \sum_{k=0}^{24} P(Y=k)$$

$$= \sum_{k=0}^{24} \frac{e^{-19.2}(19.2)^k}{k!}$$

$$= e^{-19.2}\left[1 + 19.2 + \frac{19.2^2}{2!} + \ldots + \frac{19.2^{24}}{24!}\right]$$

$$\approx 0.8842$$

Hence, $P(Y \leq 24) \approx 0.8842$

3. every 15min 中 $N \sim Poi(4.8)$

$$p = P(N \geq 5) = 1 - P(N \leq 4)$$

$$= 1 - \sum_{k=0}^{4} \frac{e^{-4.8}(4.8)^k}{k!}$$

$$= 1 - e^{-4.8}\left[1 + 4.8 + \frac{4.8^2}{2!} + \frac{4.8^3}{3!} + \frac{4.8^4}{4!}\right]$$

$$\approx 0.52374$$

~~every~~ 15min $N \geq 5$ means success

the number of success K:

$$K \sim Bin(4, p)$$

at least 3: $K \geq 3$

$$P(K \geq 3) = P(K=3) + P(K=4)$$

$$P(K=k) = \binom{n}{k} p^k (1-p)^{n-k}$$

$$P(K=3) = \binom{4}{3} p^3 (1-p)$$

$$= 4p^3(1-p)$$

$$P(K=4) = p^4$$

$$P(K \geq 3) = 4p^3(1-p) + p^4 \approx 0.3489$$

Hence, $P(K \geq 3) \approx 0.3489$

4.

```
ev = read.csv("ev.arrivals.csv")

x = ev$arrivals

mean(x)

var(x)

# count how many times each arrival count occurs

obs.counts = table(x)

obs.values = as.numeric(names(obs.counts))

emp.prop = as.numeric(obs.counts / length(x))

# P(N = k), = dpois(k, lambda)

poi.prob = dpois(obs.values, lambda = 4.8)

# make a table

comparison = data.frame(

   arrivals = obs.values,

   empirical = emp.prop,

   poisson = poi.prob)

comparison

# plot: X:arrival quantity, Y: proportion and probability

plot(obs.values, emp.prop,

        type = "h",

        lwd = 3,

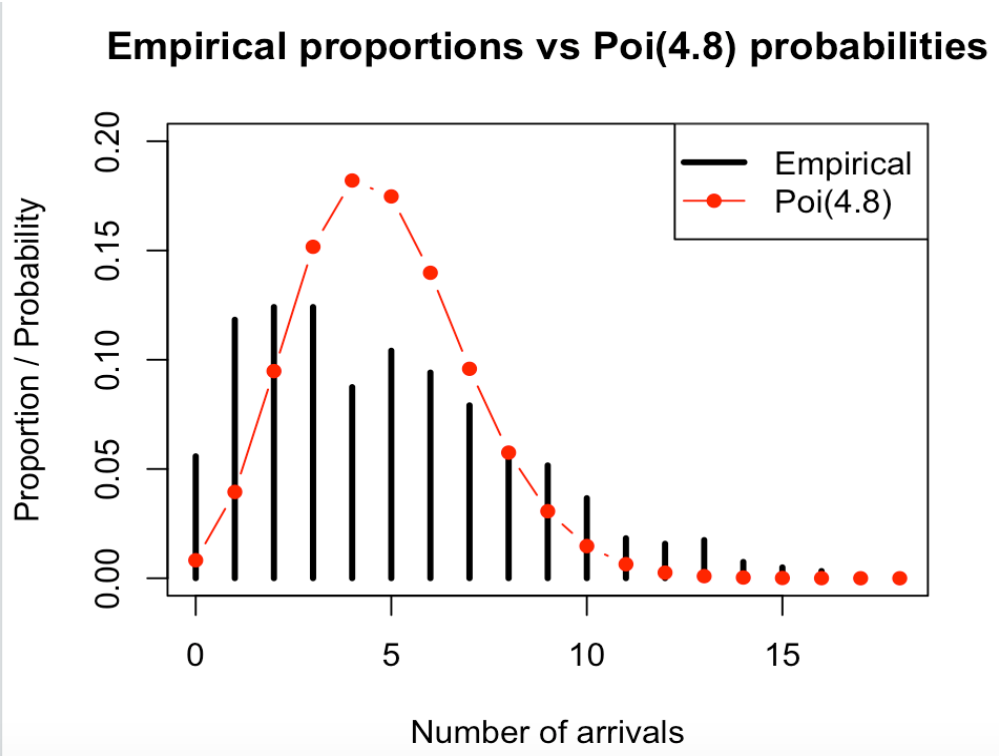
        col = "black",
```

```
        xlab = "Number of arrivals",
        ylab = "Proportion / Probability",
        ylim = c(0, 0.20),
        main = "Empirical proportions vs Poi(4.8) probabilities")
# add the points predicted by the Poisson model
points(obs.values, poi.prob,
        type = "b",
        col = "red",
        pch = 16)
# add examples
legend("topright",
        legend = c("Empirical", "Poi(4.8)"),
        col = c("black", "red"),
        lty = c(1, 1),
        pch = c(NA, 16),
        lwd = c(3, 1))
```

The Poi(4.8) model does not provide a good fit to the data. Although the sample mean, 4.8825, is close to the model mean of 4.8, the sample variance, 11.9637, is much larger than 4.8. The empirical distribution also shows substantially more high-count observations than predicted by the Poi(4.8) model. This suggests overdispersion relative to the Poisson model.

```
> mean(x)
[1] 4.8825
> var(x)
[1] 11.96366
```

The variance-to-mean ratio is approximately 2.45, whereas a Poisson model would suggest a ratio close to 1.



```r
#4.
ev = read.csv("ev.arrivals.csv")
x = ev$arrivals
mean(x)
var(x)

# count how many times each arrival count occurs
obs.counts = table(x)
obs.values = as.numeric(names(obs.counts))
emp.prop = as.numeric(obs.counts / length(x))
# P(N = k), = dpois(k, lambda)
poi.prob = dpois(obs.values, lambda = 4.8)
# make a table
comparison = data.frame(
  arrivals = obs.values,
  empirical = emp.prop,
  poisson = poi.prob)
comparison

# plot: X:arrival quantity, Y: proportion and probability
plot(obs.values, emp.prop,
     type = "h",
     lwd = 3,
     col = "black",
     xlab = "Number of arrivals",
     ylab = "Proportion / Probability",
     ylim = c(0, 0.20),
     main = "Empirical proportions vs Poi(4.8) probabilities")
# add the points predicted by the Poisson model
points(obs.values, poi.prob,
       type = "b",
       col = "red",
       pch = 16)
# add examples
legend("topright",
       legend = c("Empirical", "Poi(4.8)"),
       col = c("black", "red"),
       lty = c(1, 1),
       pch = c(NA, 16),
       lwd = c(3, 1))
# The Poi(4.8) model does not a good. Although the sample mean 4.88 is close to 4.8,
# the sample variance 11.96 is much larger than 4.8.
# The empirical distribution also has more high-count observations than predicted by the Poi(4.8) model.
```

5.

y = c(2, 3, 5, 6, 9, 8, 5, 4, 2, 1)

lambda.A = rep(4.8, 10)

lambda.B = c(2.0, 3.0, 4.5, 6.5, 8.0, 7.5, 5.5, 4.0, 2.5, 1.5)

nll.A = -sum(dpois(y, lambda.A, log = TRUE))

nll.B = -sum(dpois(y, lambda.B, log = TRUE))

nll.A # 23.32906

nll.B # 16.34228

p(yt | λt) = (e−λt λtyt)/ yt! therefore -logp(yt | λt) = λt – ytlog(λt) + log(yt!)

Interval t	Observed y_t	Model A λ_t	Model A NLL	Model B λ_t	Model B NLL
1	2	4.8	2.3559	2.0	1.3069
2	3	4.8	1.8859	3.0	1.4959
3	5	4.8	1.7444	4.5	1.7671
4	6	4.8	1.9676	6.5	1.8484
5	9	4.8	3.4843	8.0	2.0869
6	8	4.8	2.8557	7.5	1.9854
7	5	4.8	1.7444	5.5	1.7638
8	4	4.8	1.7036	4.0	1.6329
9	2	4.8	2.3559	2.5	1.3606
10	1	4.8	3.2314	1.5	1.0945
Total NLL			**23.3291**		**16.3423**

Since a lower negative log-likelihood indicates that the model assigns a higher probability to the observed data, Model B provides the better probabilistic forecasts. Model B is also able to vary its forecast λt across intervals, allowing it to better reflect changes in the observed arrival counts, whereas Model A uses the same forecast λt for every interval.

```
> nll.A
[1] 23.32906
> nll.B
[1] 16.34228
```

```r
#5.
y = c(2, 3, 5, 6, 9, 8, 5, 4, 2, 1)

lambda.A = rep(4.8, 10)
lambda.B = c(2.0, 3.0, 4.5, 6.5, 8.0, 7.5, 5.5, 4.0, 2.5, 1.5)

nll.A = -sum(dpois(y, lambda.A, log = TRUE))
nll.B = -sum(dpois(y, lambda.B, log = TRUE))
nll.A
nll.B
# Model A negative log-likelihood = 23.32906
# Model B negative log-likelihood = 16.34228
# Model B < Model A, so Model B is better.
```